

DB Health Portal · Non-Exadata Edition

User Guide

How to configure, use and interpret the DB Health Portal

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Prepared by	yuvashanthi

All screenshots in this document were captured from the real application running against **fictitious demo data** (databases PRODFIN, PRODHR, PRODCRM, PRODERP, PRODSCM, PRODBI on hosts 10.20.30.41–43). No production host names, users or credentials are shown.

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1. Introduction

The **DB Health Portal** is a Windows desktop application that opens in your web browser and shows the live health of many Oracle databases on standard (non-Exadata) servers in one place. With it you can:

- See every database's status at a glance (Healthy, Warning, Critical, Unknown).
- Check the database, the listener, RMAN backups, tablespaces, sessions and an optional reporting (standby/replica) database.
- View host CPU, RAM, top processes and mount-point usage for every server.
- Open a detailed page per database: sessions, memory, tablespaces, backups, blocking sessions, alert-log errors, growth trends, top SQL and storage.
- Receive email alerts when a problem starts, and review all alerts sent.

2. Before you start

Requirement	Details
Workstation	Windows 10/11 64-bit with a modern browser (Chrome or Edge).
Network	Your PC must reach every database host on the listener port (usually 1521) and, for OS metrics, on SSH port 22.
Database account	An Oracle account that can read the <code>V\$</code> and <code>DBA_</code> views (for example SYSTEM, or a monitoring user with SELECT_CATALOG_ROLE). The user name <code>sys</code> connects as SYSDBA.
Host account (optional)	An OS user and password on each server (usually <code>oracle</code>). It is needed for host CPU/RAM, top processes, mount points and the Archive Log Clear action.
Registry file	A text, CSV or Excel file listing your databases (section 4).
SMTP account (optional)	Needed only for email alerts.

3. Starting and stopping the portal

1. Open the `DB_Health_Portal_new` folder and double-click **DB_Health_Portal.exe**.

2. A console window opens and the portal starts a local web server. Your browser opens at `http://localhost:8501` ; if it does not, open that address yourself.
3. Keep the console window open while you use the portal.

To stop the portal, click the red  **Shutdown** button at the top right. The browser tab closes and the background process ends.

If you close every browser tab, the portal shuts itself down after about 10 seconds. To keep it running, leave at least one tab open.

4. Preparing the registry file

The registry file lists the databases to monitor, one per line. Supported formats are `.txt` , `.csv` , `.xlsx` and `.xls` . In a `.txt` file, columns are separated by spaces and read **by position**.

#	Column	Required	Meaning
1	db_name	Yes	Name shown in the portal. Use <code>NC</code> if this line has only a reporting database.
2	host	Yes	Database server IP address or host name.
3	port	Yes	Listener port, usually 1521.
4	service_name	Yes	Service name. If connecting with the service name fails, the portal tries it as a SID.
5	username	Yes	Oracle user (<code>sys</code> connects as <code>SYSDBA</code>).
6	password	Yes	Oracle password.
7	host_username	Optional	OS / SSH user (usually <code>oracle</code>).
8	host_password	Optional	OS / SSH password. Needed for OS metrics, processes and mount points.
9	reporting_db_name	Optional	Name of the reporting database. <code>NC</code> means none.
10–14	reporting_host, reporting_port, reporting_service_name, reporting_username, reporting_password	Optional	Connection details for the reporting database.

Demo registry used for this guide (all values are fictitious):

```
# db_name host port service_name username password host_username host_password reporting_db ...
PRODFIN 10.20.30.41 1521 PRODFIN system ***** oracle ***** RPTFIN 10.20.30.51 1521 RPTFIN r
pt_user *****
PRODHR 10.20.30.41 1521 PRODHR system ***** oracle ***** NC
PRODCRM 10.20.30.42 1521 PRODCRM system ***** oracle ***** RPTCRM 10.20.30.52 1521 RPTCRM r
pt_user *****
PRODERP 10.20.30.42 1521 PRODERP system ***** oracle ***** NC
PRODSCM 10.20.30.43 1521 PRODSCM system ***** oracle ***** NC
PRODBI 10.20.30.43 1521 PRODBI system ***** oracle ***** NC
```

Rules for .txt files

- Start comment and header lines with #. A header without # is read as a database called "db_name".
- Values must not contain spaces.
- To leave a column empty but still fill a later one, write - (for example, no SSH credentials but a reporting database).
- CSV and Excel files may have a header row; it is detected automatically.

The registry file holds passwords in plain text. Store it in a folder only DBAs can read, and never email it or put it in source control.

5. Registry setup screen

Each time you open the portal in a new browser tab (or refresh the page), the **Database Registry Setup** screen appears first.

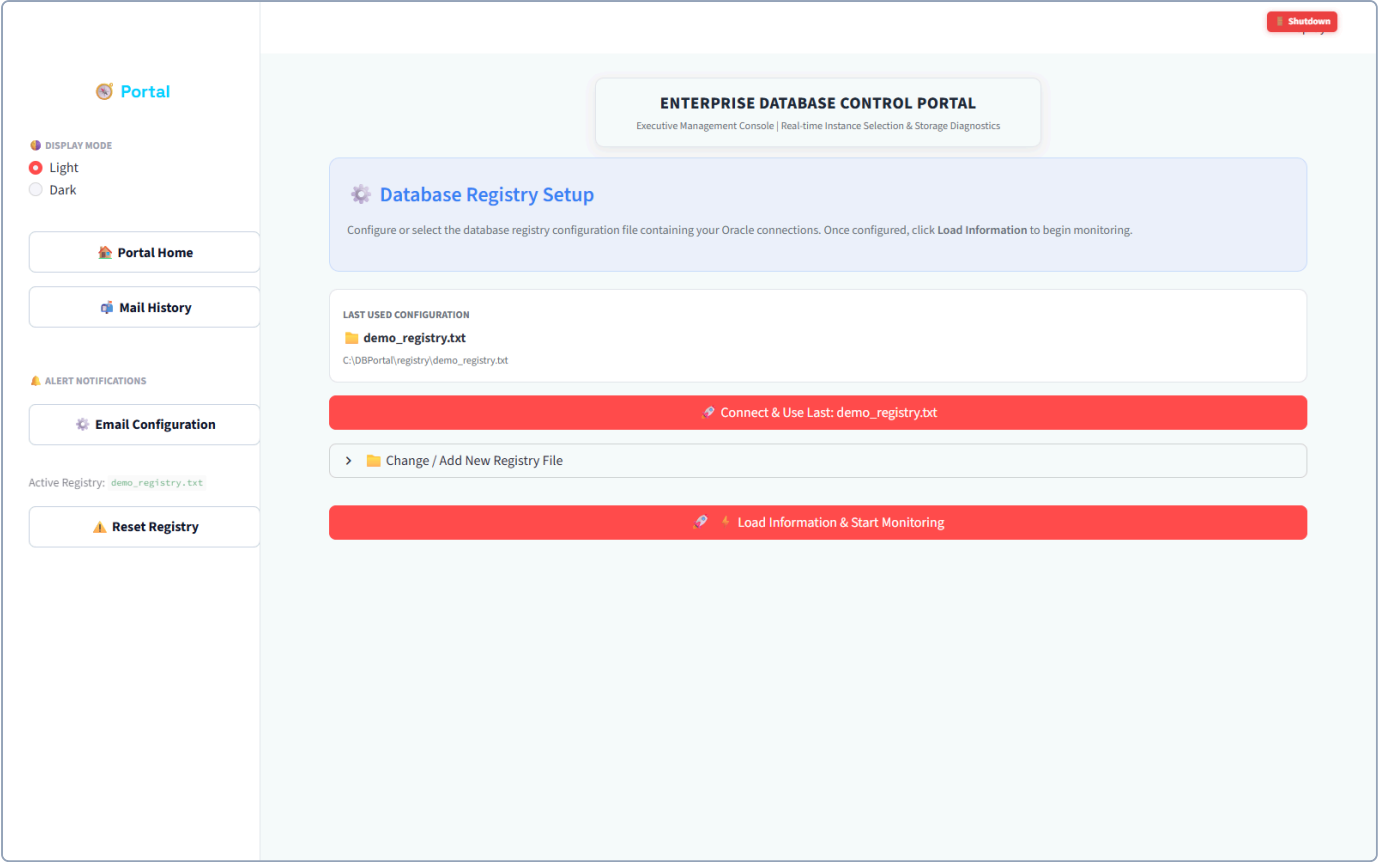


Figure 1: Database Registry Setup screen with the last-used registry (demo data)

Control	What it does
LAST USED CONFIGURATION	Shows the registry file used last time and its full path.
Connect & Use Last: <file>	Opens the dashboard straight away with that file.
Change / Add New Registry File	Expands so you can upload a different file or type a path.
Load Information & Start Monitoring	Starts monitoring with the active registry and opens Portal Home.

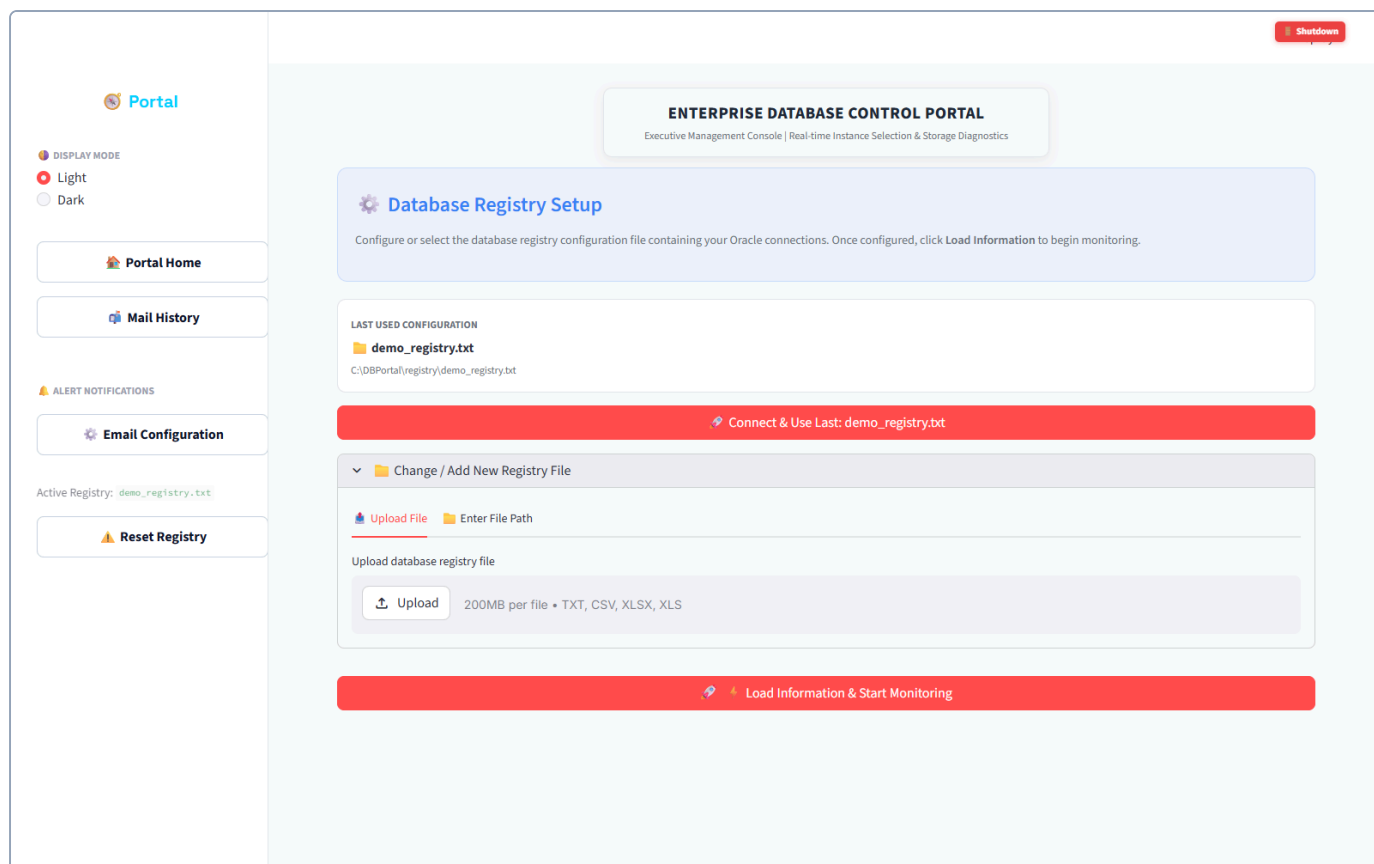


Figure 2: Change / Add New Registry File – Upload File tab (demo data)

Upload File: drag a .txt/.csv/.xlsx/.xls file onto the box or click *Browse files*. The file is copied into the portal's `scratch` folder and becomes the active registry, and the message *Registry file uploaded* confirms it.

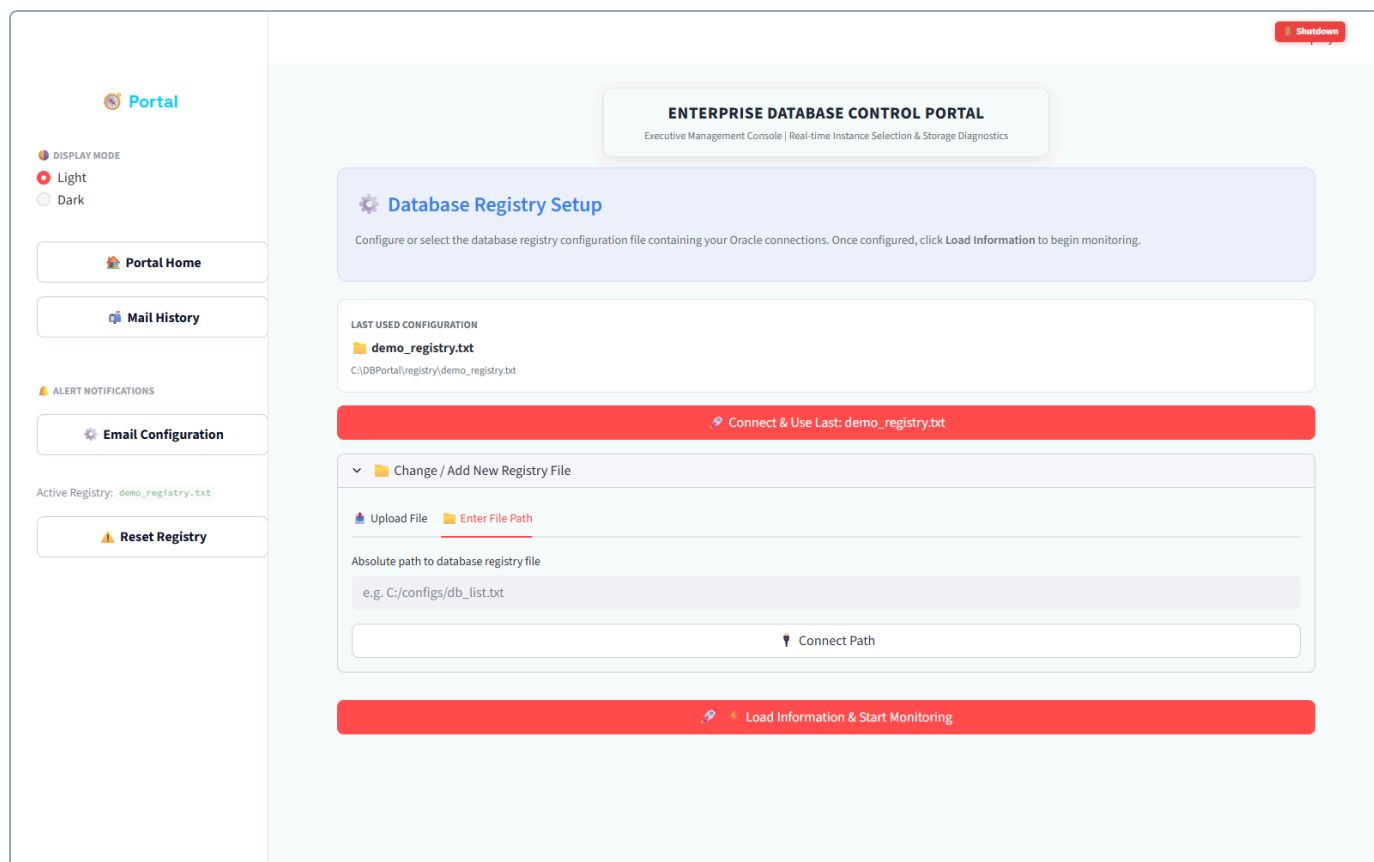


Figure 3: Change / Add New Registry File – Enter File Path tab (demo data)

Enter File Path: type the full path, for example `C:/configs/db_list.txt` , and click  **Connect Path**.

If the path does not exist, nothing happens and no error message appears. Check the path carefully.

6. Portal Home (overview)

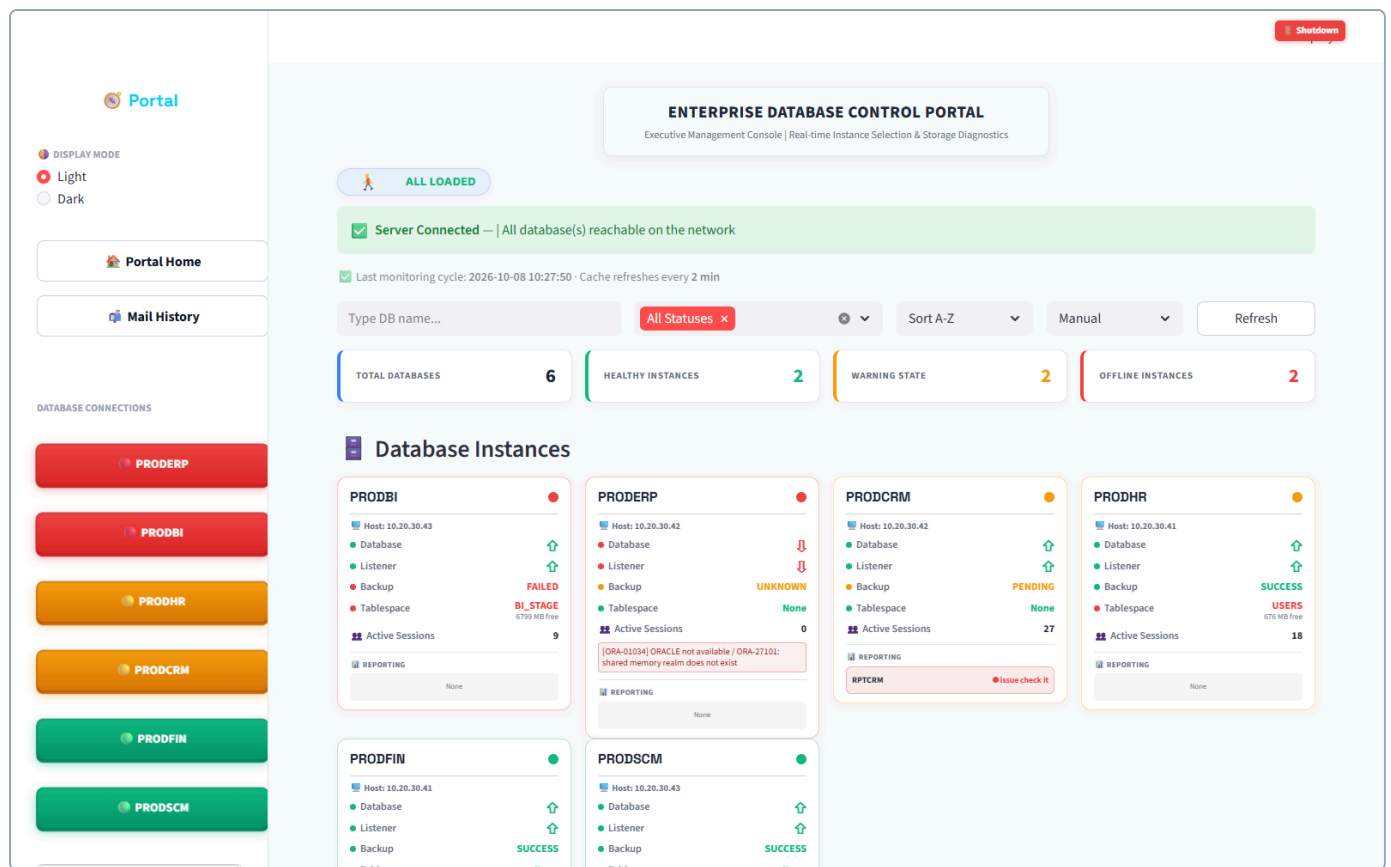


Figure 4: Portal Home – status ribbon, filters, KPI tiles and database cards (demo data)

From top to bottom, the page shows:

1. **Loading pill:** shows *RUNNING...* while statuses are being collected and *ALL LOADED* when finished.
2. **Connection banner** and **Last monitoring cycle:** the time of the latest background check. The background cache refreshes every 2 minutes.
3. **Controls row:**
 - **Search** (*Type DB name...*): filters the cards by name.
 - **Status filter:** All Statuses, Healthy, Warning or Critical (Figure 6).
 - **Sort:** A-Z or Z-A.
 - **Refresh mode:** Manual or Automatic. In Automatic mode, set *Refresh every N minute(s)*; the default is 20.
 - **Refresh:** clears the cached statuses and checks every database again.
4. **KPI tiles:** Total Databases, Healthy Instances, Warning State and Offline Instances.
5. **Database Instances:** one card per database, 4 per row, with the worst status first.
6. **Host Server Resources & Diagnostics:** one block per server (section 9).

Offline Instances counts every database in **Critical** state. That includes databases that are up but whose last backup failed or whose listener is down. Databases in **Unknown** state appear only in the total.

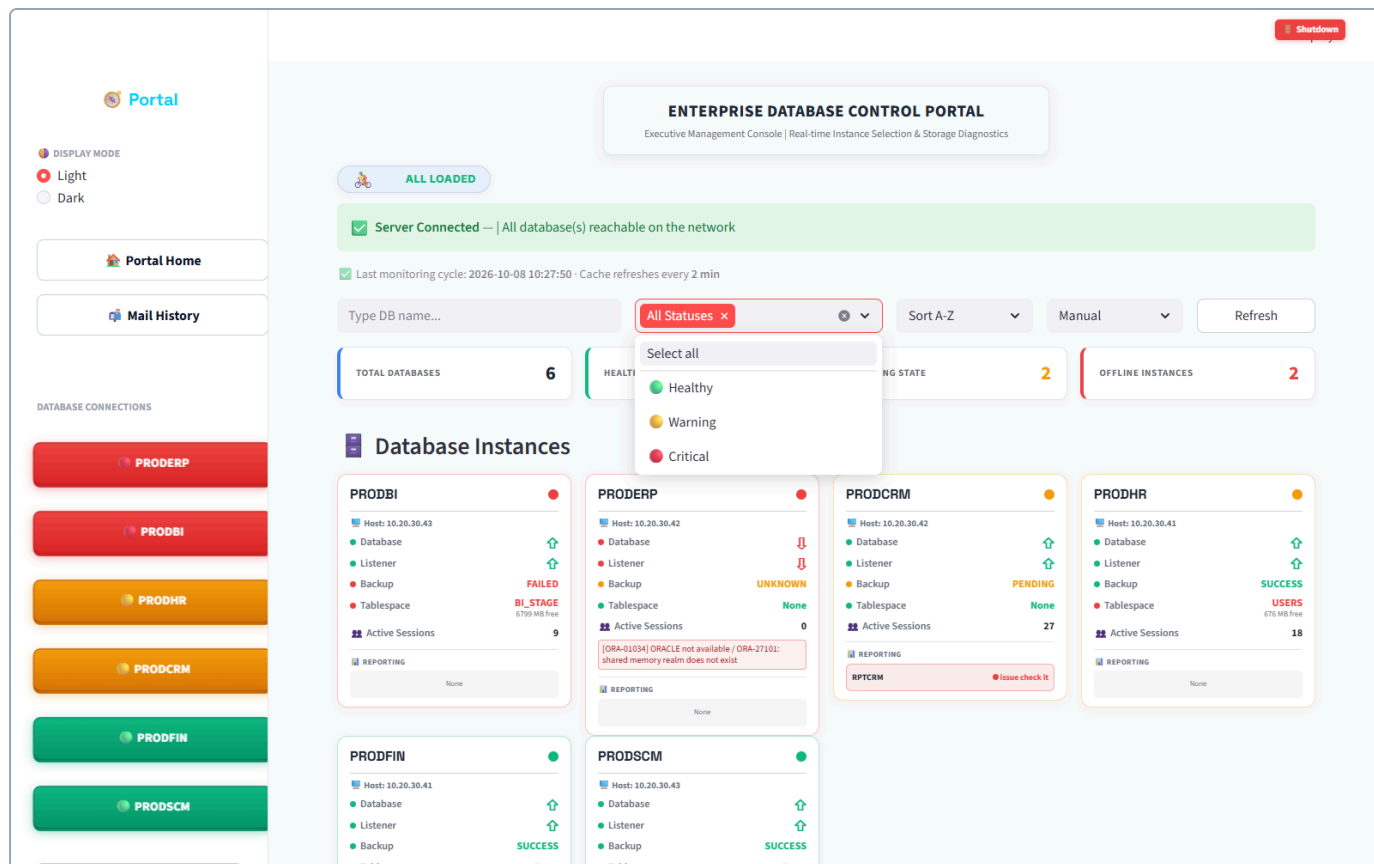


Figure 5: Status filter options (demo data)

Choosing a specific status hides Unknown databases. "All Statuses" cannot be combined with the other options.

7. Database cards

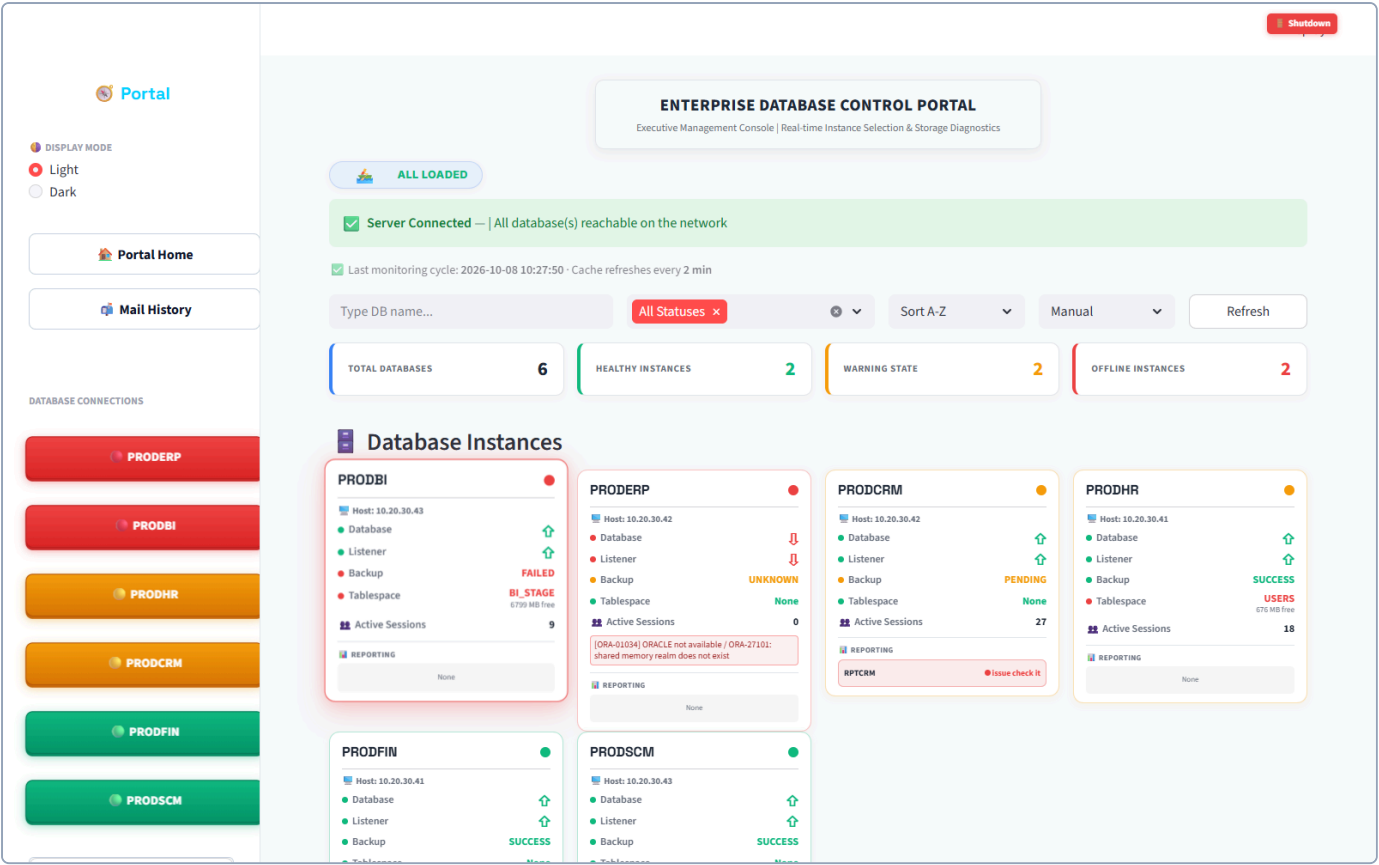


Figure 6: Cards for Critical (PRODBI, PRODERP), Warning (PRODCRM, PRODHR) and Healthy databases (demo data)

Row	Values and colours
Header dot	Health colour. Hover over it to see <i>System Diagnostics</i> , the list of reasons for the status.
Host	Server the database runs on.
Database	⬆ green = OPEN; ⬇ red = DOWN; ⬆ amber = other states.
Listener	⬆ up / ⬇ down.
Backup	Result of the latest RMAN job in the last 24 hours: SUCCESS FAILED PENDING (no job in the last 24 h, or still running) UNKNOWN (could not be read).
Tablespace	<i>None</i> if every permanent tablespace is below 90%. Otherwise, each tablespace at 90% or more, in red, with the MB free.
Active Sessions	Number of active user sessions.
Error box	The connection error text when the database cannot be reached (for example ORA-01034).
REPORTING	The reporting database name with UP (green) or issue check it (red). <i>None</i> means no reporting database is configured.

Click a card, or its name in the sidebar, to open the database detail page (section 11).

8. Health status rules

The rules are checked in this order; the first that matches decides the status.

Status	Condition
Unknown	The status has not been collected yet, or the line is <code>NC</code> (not configured).
Critical	Database DOWN, or listener DOWN, or last backup FAILED.
Warning	Backup PENDING, or a permanent tablespace at 90% or more, or (when the background monitor has collected them) blocking sessions, a deadlock, or sessions near the limit.
Healthy	None of the above.

In the demo: PRODERP is Critical (database and listener down), PRODBI is Critical (backup failed), PRODHR is Warning (USERS tablespace at 93.4%), PRODCRM is Warning (backup pending), and PRODFIN and PRODSCM are Healthy.

9. Host server resources

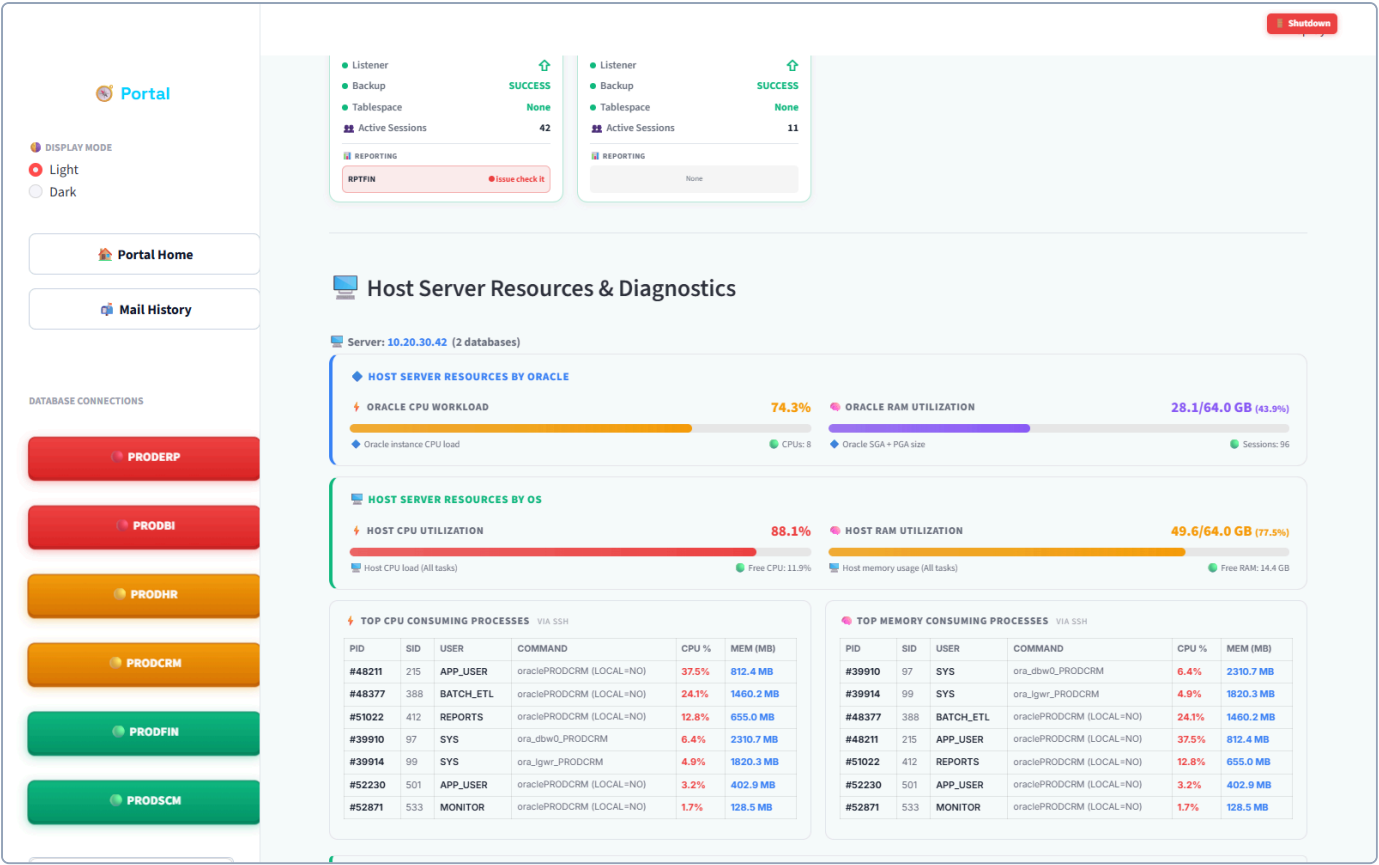


Figure 7: Host section – Oracle and OS resource cards (demo data)

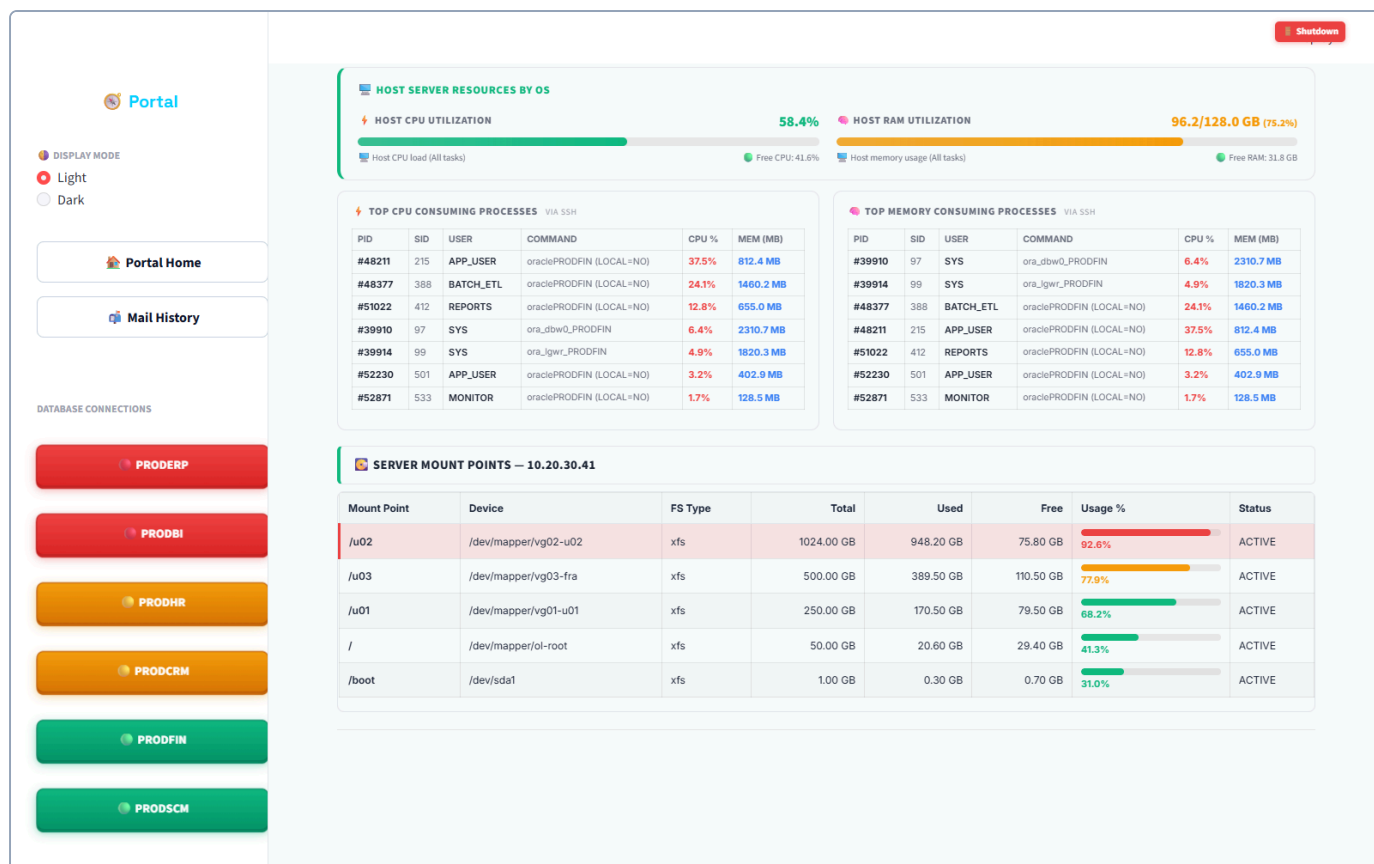


Figure 8: Top processes and server mount points for host 10.20.30.41 (demo data)

Each server appears as **Server: <host> (N databases)** with these blocks:

- **Host Server Resources by Oracle:** Oracle CPU workload (%) and Oracle RAM (SGA + PGA) against total RAM, plus the CPU count and session count.
- **Host Server Resources by OS:** whole-server CPU and RAM use, from SSH when a host password is set, otherwise from Oracle's OS statistics.
- Colours for both resource cards: red above 85%, amber above 70%.
- **Top CPU / Top Memory Consuming Processes:** the 10 heaviest Oracle processes. This table refreshes on its own every 30 seconds.
- **Server Mount Points:** every filesystem with size, used, free and usage %. Rows above 90% are highlighted red; amber means above 75%.

If every database on a server is down, the block shows *Server resources offline or unreachable*.

10. Sidebar navigation

Item	Purpose
Display Mode	Switch between the Light and Dark themes.
Portal Home	Go back to the overview page.
Mail History	Open the email alert log (section 13).
Database Connections	One coloured button per database, worst first (red, amber, green, grey). (Active) marks the open one; checking... means its status is still loading.

 Previous Registries	Switch quickly to a registry file used before.
 Email Configuration	SMTP and alert settings (section 12).
 Reset Registry	Unload the current registry and return to the setup screen.

11. Database detail (Monitoring) page

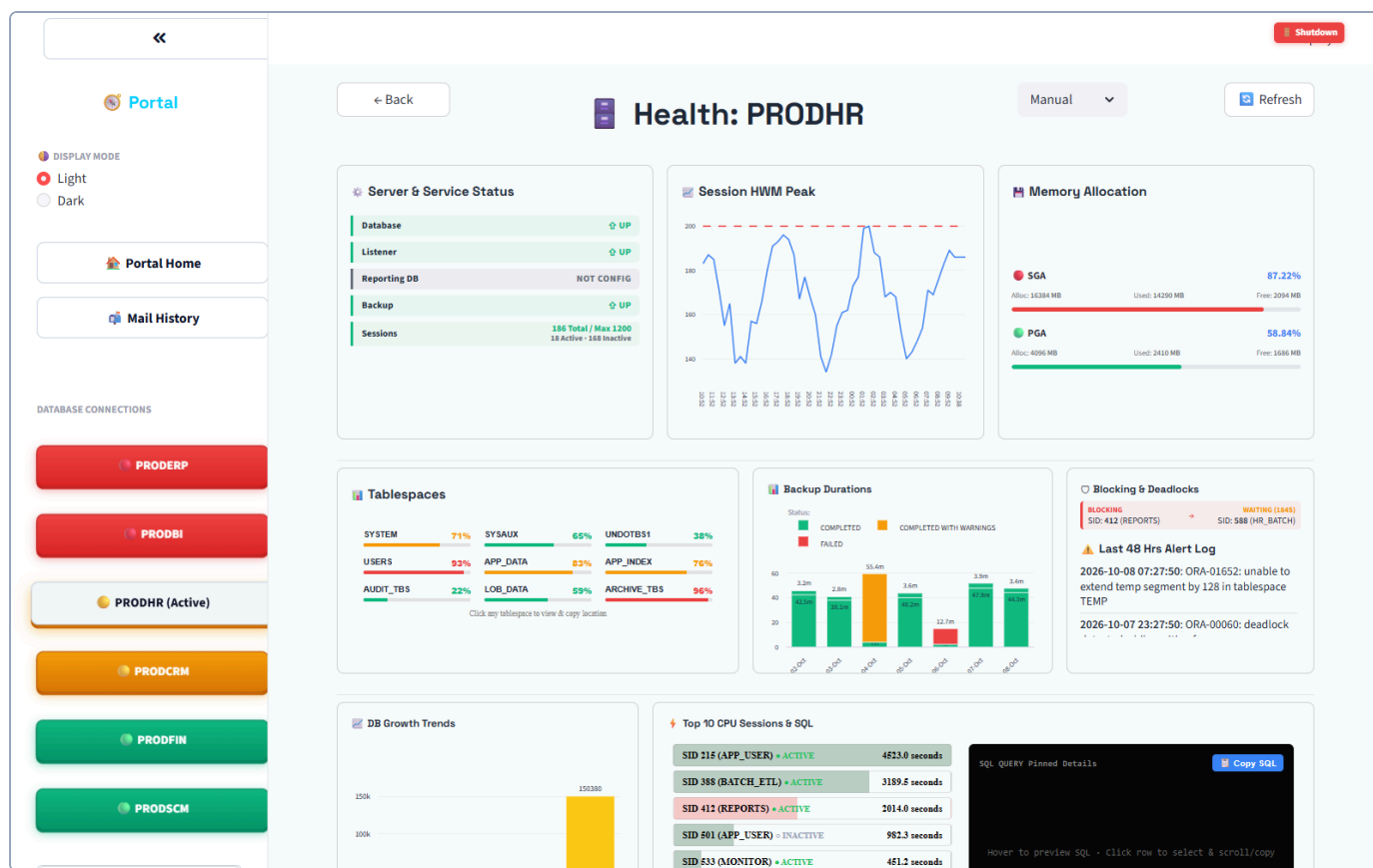


Figure 9: Health: PRODHR – status, sessions, memory, tablespaces, backups and blocking (demo data)

11.1 Header

← **Back** returns to Portal Home. Next to it are the refresh mode (Manual / Automatic) and **Refresh**, which reloads every panel. Data is cached for 60 seconds, so switching theme does not trigger a new query.

11.2 Panels

Panel	What it shows	Colour rules
Server & Service Status	Database, Listener, Reporting DB and Backup status; total, active and inactive sessions against the session limit.	Sessions: red when active sessions exceed the larger of 50 or 85% of the limit; amber when they exceed the larger of 20 or 60%.
Session HWM Peak	Line chart of sessions over the last 24 hours. The red dashed line is the 24-hour peak.	–
Memory Allocation	SGA and PGA allocated, used and free (MB) and % used.	Red above 85%.
Tablespaces	Up to 12 tablespaces, each with % used against its maximum size. Click a name to see and copy its datafile location.	Red above 85%, amber above 70%.
Backup Durations	RMAN jobs over the last 7 days, in minutes.	Green completed; amber completed with warnings; red failed; blue running.

Blocking & Deadlocks	Each blocker → waiter pair with SIDs, users and wait time.	–
Last 48 Hrs Alert Log	ORA- errors from the alert log in the last 2 days (up to 50).	–
DB Growth Trends	Daily, weekly, monthly and yearly growth (MB).	Based on MB per day: green up to 100, yellow up to 500, orange up to 1000, red above.
Top 10 CPU Sessions & SQL	Sessions using the most CPU. Hover over a row to preview its SQL; click a row to pin it and use Copy SQL .	Red rows are sessions involved in blocking.
Archive Log Destination	Fast Recovery Area (FRA) or custom destination: allocated, used and free (GB).	Red at 90% or more, amber at 70% or more.
Top Running Processes	CPU and memory heavy sessions of this database. Refreshes every 30 seconds.	–
OS Storage / Mount Points	Filesystems and ASM disk groups (+DATA).	Red at 90% or more, amber at 70% or more.

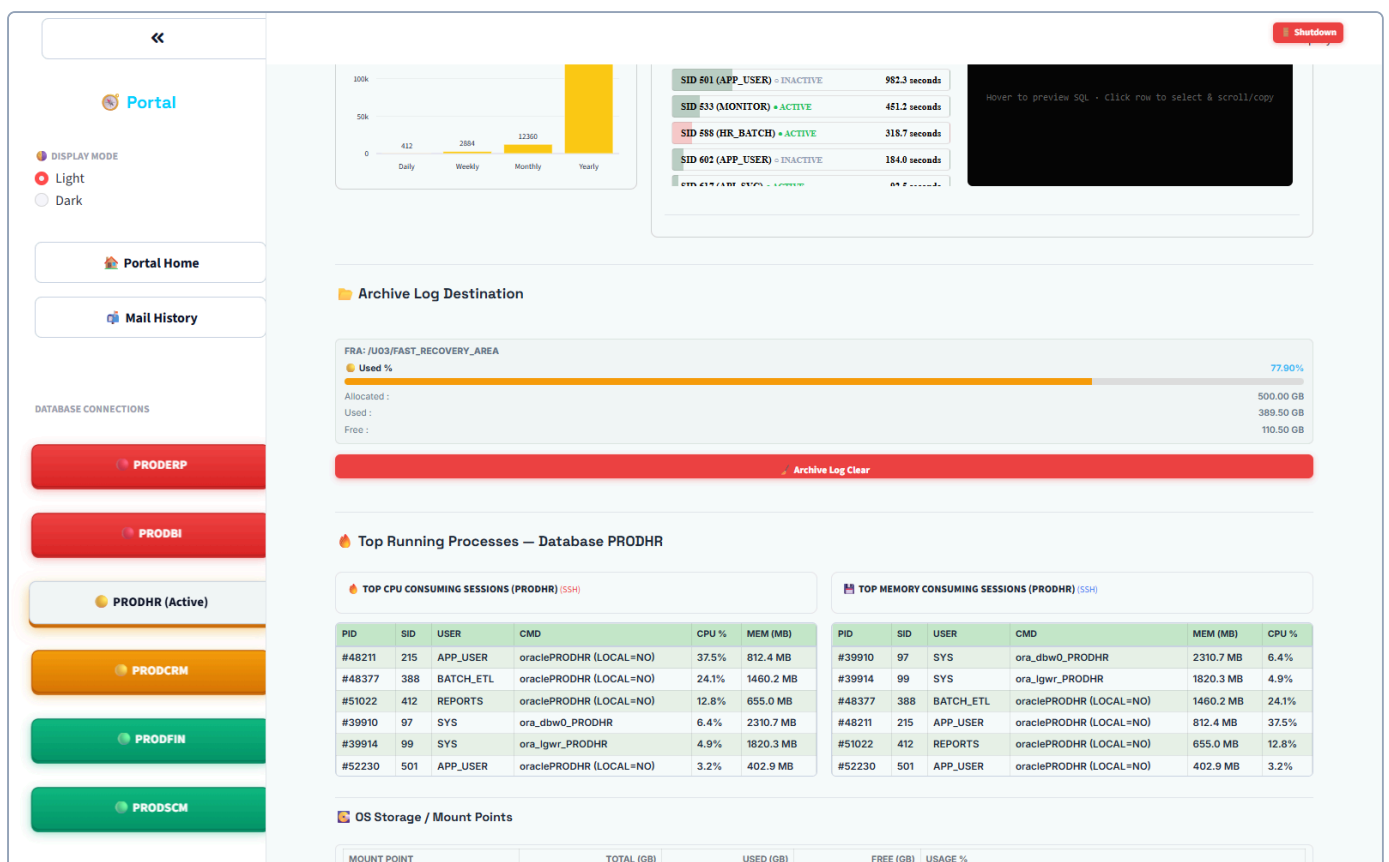


Figure 10: Growth trends, top CPU sessions and archive log destination (demo data)

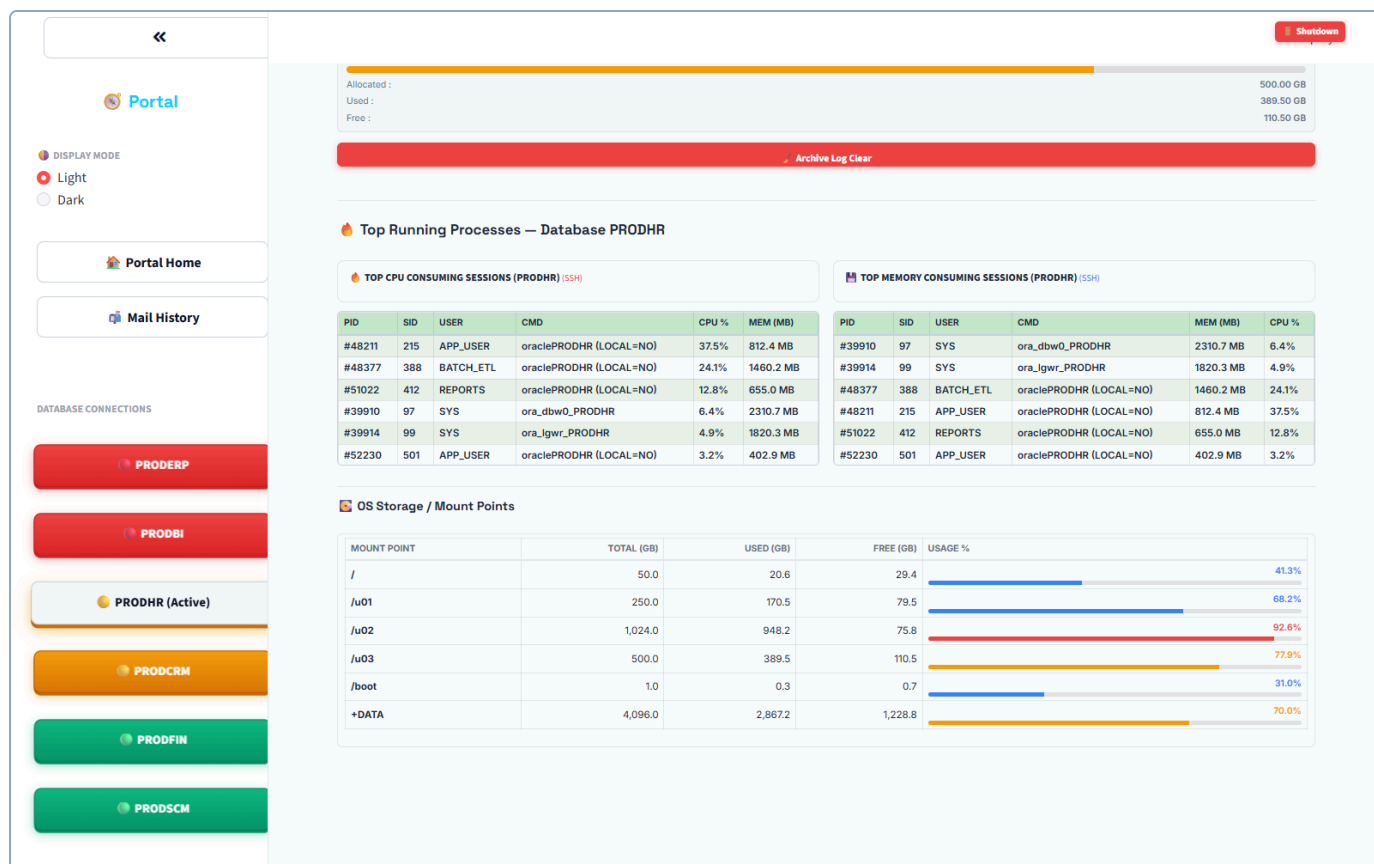


Figure 11: Top running processes and OS storage / mount points (demo data)

Actions that change the database

- **Kill Session** (from the CPU sessions panel) runs `ALTER SYSTEM KILL SESSION ... IMMEDIATE`.
- **Archive Log Clear** connects over SSH and runs `RMAN DELETE NOPROMPT ARCHIVELOG ALL COMPLETED BEFORE 'SYSDATE-15'`.

Neither action asks for confirmation. Use them only when authorised, and make sure archive logs are backed up first.

12. Email configuration

DATABASE CONNECTIONS

PRODERP

PRODBI

PRODHR

PRODCRM

PRODFIN

PRODSCM

> Previous Registries

ALERT NOTIFICATIONS

Email Configuration

Active Registry: demo_registry.txt

Reset Registry

Shutdown

Email Configuration

Configure SMTP settings, test email delivery, and choose which alert events trigger notifications.

SMTP Server Settings

SMTP Server

smtp.demo-corp.local

SMTP Port

587

SMTP Username

dba-alerts@demo-corp.local

SMTP Password

Sender Email Address

dba-alerts@demo-corp.local

Recipient Email Address(es)

dba-team@demo-corp.local, oncall@demo-corp.local

Enable Email Alerts

Save Configuration

Send Test Email

Notification Preferences

Select the alert types for which you want to receive email notifications. Deselected alerts will be suppressed – no email will be sent, but events are still logged.

Select All

Deselect All

Figure 12: Email Configuration – SMTP server settings (demo data)

Field	Description
SMTP Server / SMTP Port	Mail relay and port: 25 (relay), 587 (STARTTLS) or 465 (SSL).
SMTP Username / SMTP Password	Account used to log in to the mail server.
Sender Email Address	The "From" address on alert emails.
Recipient Email Address(es)	Comma-separated list of recipients.
☒ Enable Email Alerts	Master switch. When it is off, alerts are only logged (status <i>Disabled</i> in Mail History).

Click  **Save Configuration** to store the settings, and  **Send Test Email** to check delivery. Settings are saved per registry file.

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DATABASE CONNECTIONS

PRODERP
PRODBI
PRODHR
PRODCRM
PRODFIN
PRODSCHM

> Previous Registries

ALERT NOTIFICATIONS

Email Configuration

Active Registry: demo_registry.txt

Reset Registry

DATABASE AVAILABILITY

☒ Database Down
☐ Database Instance Restarted
☒ Listener Down
☐ Host Server Down
☐ Database Not Reachable

BACKUP

☐ Backup Failure

TABLESPACE & DATAFILES

☒ Tablespace Usage Critical ($\geq 90\%$)
☐ Temporary Tablespace Critical
☐ Tablespace Usage Warning ($\geq 80\%$)
☐ Datafile Autoextend Failure
☐ Undo Tablespace Critical

ARCHIVE LOG & FRA

☒ Archive Log Destination Full
☐ Archive Log Generation Failure
☐ FRA Usage Critical ($\geq 90\%$)

SESSIONS & LOCKS

☐ High Session Count ($> 85\%$ max)
☒ Deadlock Detected
☒ Blocking Sessions Detected
☐ Long Running Sessions

CPU & MEMORY

☐ High CPU Usage
☐ Memory Consuming Process Threshold
☐ High Memory Usage
☐ CPU Consuming Process Threshold

STORAGE & MOUNT POINTS

☒ Mount Point Usage Critical ($\geq 90\%$)
☐ Disk Usage Critical
☐ Mount Point Usage Warning ($\geq 80\%$)
☐ ASM Diskgroup Usage Critical
☐ Filesystem Not Accessible

Shutdown

Figure 13: Notification Preferences – choose which alert types send email (demo data)

Tick the alert types you want, or use **Select All** / **Deselect All**, then click **Save Notification Preferences**.

In this build only these 8 alert types actually send email:

- Database Down
- Listener Down
- Tablespace Usage Critical ($\geq 90\%$)
- Archive Log Destination Full
- Mount Point Usage Critical ($\geq 90\%$)
- ORA Error Detected
- Deadlock Detected
- Blocking Sessions Detected

The other check boxes are saved but have no effect yet. Thresholds are fixed and cannot be changed in the UI. An email is sent once, when a problem first appears; you receive another only after the problem clears and comes back.

13. Mail History

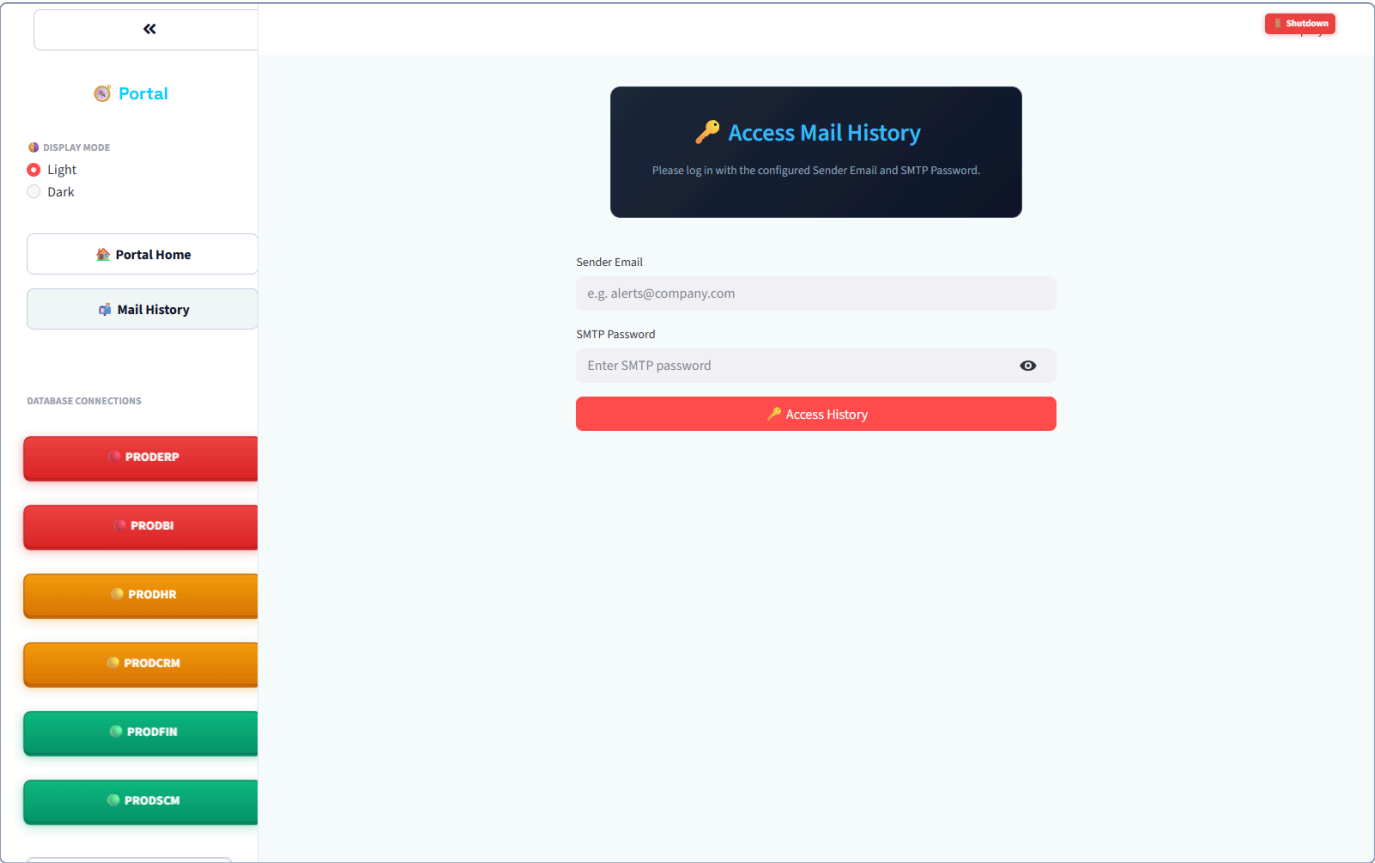


Figure 14: Mail History login (demo data)

To protect the alert log, Mail History asks for the configured **Sender Email** and **SMTP Password**.

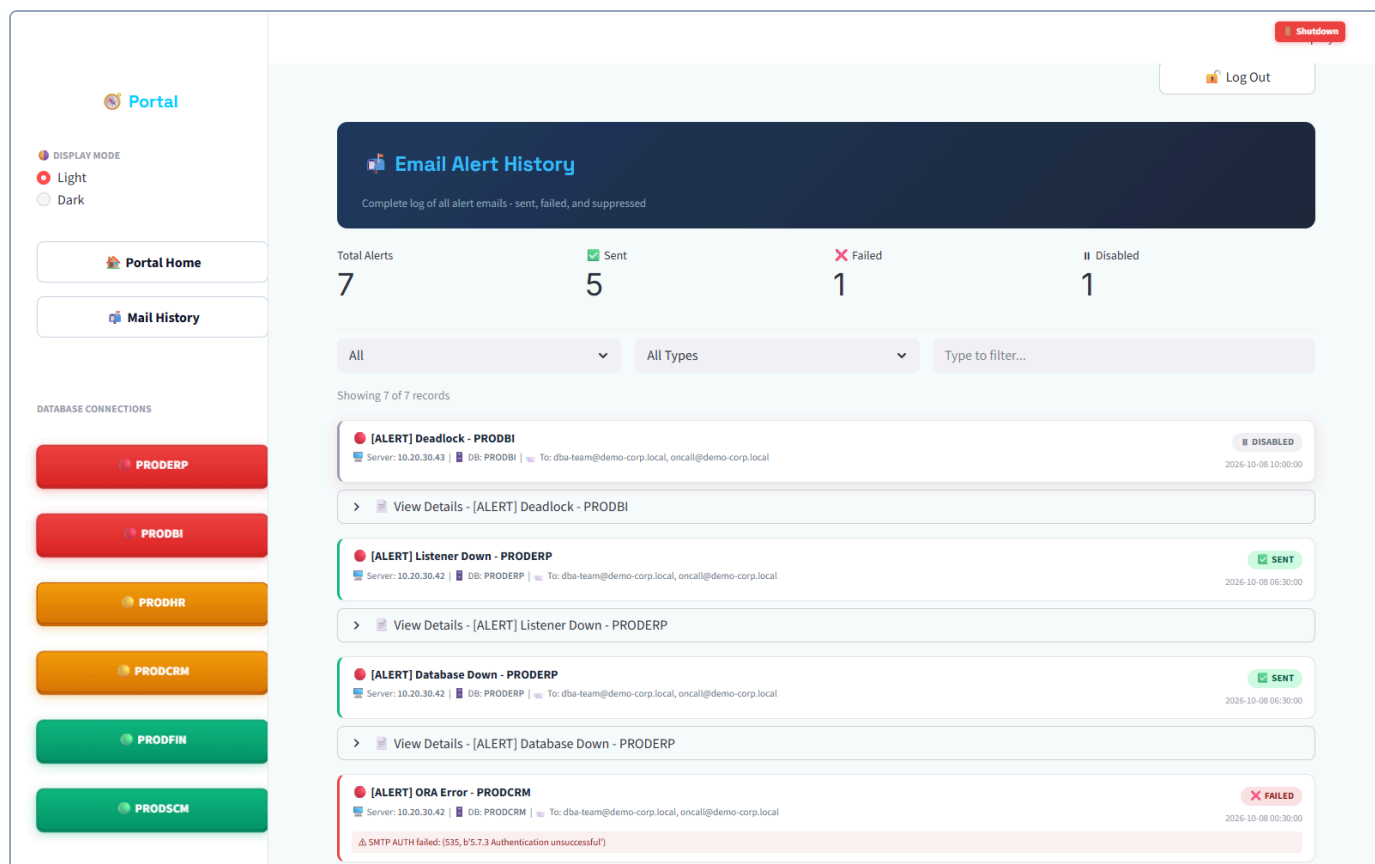


Figure 15: Email Alert History with sent, failed and disabled alerts (demo data)

- **Tiles:** total, sent, failed and disabled alerts.
- **Filters:** by status, by alert type, and a free-text search by database or server.
- **Each card** shows the subject, server, database, recipients, time and status. Failed cards also show the SMTP error.
- **View Details** shows the email body.
- **Clear All History** deletes the log permanently.
- **Log Out** locks the page again.

14. Refresh behaviour

Item	Interval
Background monitor (all databases, alert emails)	Every 2 minutes. The Refresh button forces an immediate cycle.
Sidebar database list	Every 3 seconds, from the cache.
Top-process tables (home and detail pages)	Every 30 seconds.
Detail page metrics cache	60 seconds.
Automatic mode (home and detail pages)	User-defined; default 20 minutes.

15. Troubleshooting & FAQ

Symptom	Likely cause / action
---------	-----------------------

Card shows <code>ORA-12541</code> , <code>DPY-6005</code> or a timeout	The listener cannot be reached from your PC. Check the host, the port and the firewall.
Card shows <code>ORA-01017</code>	Wrong Oracle user name or password in the registry.
✗ host username and password is wrong	Wrong <code>host_username</code> / <code>host_password</code> , or SSH password login is disabled on the server.
"No host_password configured..."	Add an OS password to at least one line for that host to get processes and mount points.
Backup always shows PENDING	No RMAN job has finished in the last 24 hours, or backups run from another tool that does not write to <code>V\$RMAN_BACKUP_JOB_DETAILS</code> .
REPORTING shows <i>issue check it</i>	The reporting database is unreachable, or its check is still running. Wait one refresh cycle before investigating.
Test email fails	Check the server, port and credentials, and that your network allows outbound SMTP. See the error text in Mail History.
Portal closed by itself	All browser tabs were closed for 10 seconds or more. Start the exe again.
Database name "db_name" appears	The header line in the .txt registry is missing its leading <code>#</code> .

16. Glossary

Term	Meaning
Registry	The file listing the databases to monitor.
Reporting DB	A secondary database (standby, replica or reporting copy) linked to a production database.
HWM	High-water mark: the highest session count in the period.
FRA	Fast Recovery Area: Oracle's disk area for archive logs and backups.
SGA / PGA	Oracle shared memory and per-process memory.
NC	"Not configured" marker in the registry.